MA1521 L1 Week 08 Mon 20261005

Evaluate each of the following iterated integrals:

a) $\displaystyle\int_0^2 \int_{-1}^1 (6x^2 + 2y)\, dy\, dx$

b) $\displaystyle\int_0^{\pi/2} \int_0^{\pi} (u \sin t + t \cos u)\, dt\, du$

Evaluate each of the following iterated integrals:

c) $\int_0^1 \int_{\sqrt{x}}^{x^2} 2x^2 y \, dy \, dx$

d) $\int_0^1 \int_0^u \sqrt{u^2+4} \, dv \, du$

Evaluate each of the following double integrals over the indicated region R.

a) $\iint_R x\,dA$; R is the finite region bounded by the axes and $2y + x = 2$

Evaluate each of the following double integrals over the indicated region R.

b) $\iint_R (2x + y^2)\, dA$; R is the finite region in the first quadrant bounded by the axes and $y^2 = 1 - x$

Evaluate each of the following iterated integrals, by changing the order of integration

a) $\displaystyle\int_0^2 \int_x^2 e^{-y^2}\, dy\, dx$

b) $\displaystyle\int_0^{1/4} \int_{\sqrt{t}}^{1/2} \frac{e^u}{u}\, du\, dt$

c) $\displaystyle\int_0^1 \int_{x^{1/3}}^1 \frac{1}{1+u^4}\, du\, dx$